

Simple facts

- Logic Friday is a free software for Boolean logic optimisation, analysis and synthesis.
- It works on Windows and needs Internet Explorer version 5 or later, on your system.
- A version for MAC obtained from Wine is included in this DVD, along with the Windows version. Make sure you have X11 on your MAC.
- You will need Espresso and MisII while working with Logic Friday.
- You can deal with up to 16 inputs and 16 outputs for a given function.

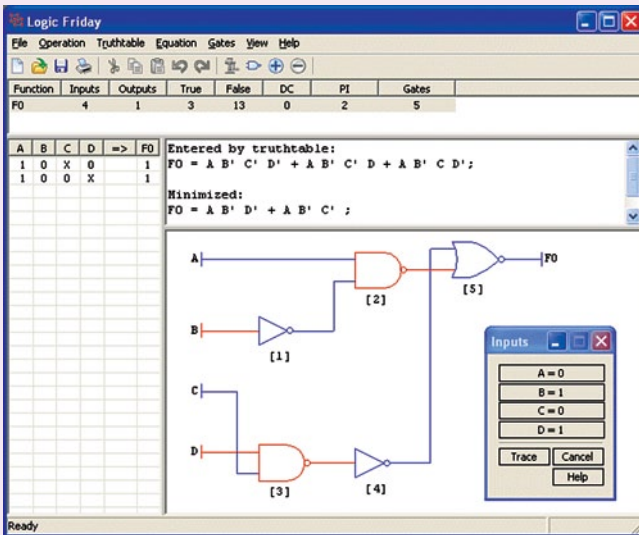


Fig. 3: Trace the logic in your gates

mapping your design to gates.

Logic Friday has support from MisII for gate diagram synthesis. You can optimise the diagram for standard integrated circuits (ICs) that will minimise the total number of IC packages in your design, or optimise for die area, letting the program choose gates to create a custom IC with minimum die area. Mapping to gates automatically minimises function.

The other important factor is tracing logic in a gate diagram. Choosing this option automatically adds colour to the diagram, with a red for logic high and blue for logic low. You can thus view how your signal travels from input to any point in the circuit. To observe the result for other cases, change the input combination in the tool window and visualise the trace displayed.

Compare functions to find differences

Logic can get very confusing. Two

and find the differences between these.

Possible cases could be equal, unequal or one being a sub-set of the other. With various input combinations possible, the tool even tells you exactly for which rows the two functions happen to be alike.

Further, you even have a choice to clone a function to save a copy. Very useful when you have to move back a few paces while designing something complex, would you not say?

Play around with your equations

Depending on need, convert your equation to look like you want it to. Be it a sum-of-products form or a product-of-sums form, factorise it or leave it expanded. The variable names and equation formats follow a given syntax of representation. The equation can be assigned values of true, false, don't care or invert. If the input to an equation is a don't care, the tool understands the input to

have various values and creates an exclusive table for it, while not considering it for minimisation. A don't care in the output is important as it influences calculation of results.

A look-up table for implementing in software

This feature comes in very useful when you have a logic function that needs to be carried forward to software or firmware. Simply choose the 'generate C look-up function' tab under operations menu and Logic Friday creates a stand-alone module that can be added to your source files. You need to specify the size of type 'unsigned int' on the processor or compiler for the software to properly distribute output truth values over bits of a data table. You can select how you want the function to be represented and called in C code. Every output will be assigned its own look-up table and don't care conditions will be considered logic high.

Offering the much needed flexibility

Once done with experimentation and designing the logic circuit, Logic Friday makes it easy to save your design by simply exporting the truth table into an Excel file with a .csv extension, and the gate diagram into an image file in formats such as Windows bitmap (.bmp) or enhanced metafile (.emf).

For logic equation or gate diagram, print these to save a copy. For diagrams that may not be legible in a single page, a road atlas style is followed to print on two sheets, without losing continuity.

You can even choose to import a truth table before you begin working on your design.

Sounds simple enough, right? Well, what are you waiting for then? Grab this month's DVD and install the tool. Explore. Have fun designing! **EFY**

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